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| TEAM ID | SWTID-2026-6767 |
| PROJECT TITLE | Visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau |
| MAX MARKS | 5 MARKS |

# CHAPTER – VII

# FUNCTIONAL AND PERFORMANCE TESTING

## 7.1 – Performance Testing & Functional Testing

**Performance Testing:**

Performance testing is used to evaluate how efficiently the Tableau dashboard handles data processing, filtering, and visualization rendering. It helps ensure that the dashboard provides timely and reliable insights without delays.

The key performance parameters considered are:

- Amount of data rendered from the database.
- Utilization of data filters.
- Number of calculation fields used in the dashboard.

**1. Amount of Data Rendered to the Database**

The amount of data rendered depends on the size of the dataset and the database's capability to store and retrieve records efficiently. Monitoring the volume of data retrieved helps ensure that database queries are optimized and do not overload the system.

**Steps to Check Table Information in MySQL Workbench**

1. Open **MySQL Workbench**.
2. Select the required database.
3. Expand the **Tables** section.
4. Select the desired table.
5. Click the **(i) Information** button.
6. View table details such as:

- Number of columns.
- Number of rows.
- Data types.
- Storage information.

Monitoring these properties helps in estimating the amount of data processed by Tableau and improving query performance.

**2. Utilization of Data Filters**

Efficient use of filters reduces the volume of data processed and displayed, thereby improving dashboard responsiveness and enabling users to perform focused analysis.

**Dashboard 1 Filters**

The following filters are available on the right-hand side of the dashboard:

**Body Style Filter**

Allows users to analyse EV cars based on body type, such as:

- Cabriole
- Hatchback
- Lift back
- MPV
- Pickup
- Sedan
- SPV
- Station Wagon
- SUV

**Brand Filter**

Allows users to filter data by EV manufacturers, including:

- Audi
- BMW
- Tesla
- Ford
- Jaguar
- Hyundai
- Tata
- Mercedes-Benz

- Porsche
- Volvo
- Nissan, etc.

These filters improve performance by displaying only the selected subset of records.

**Dashboard 2 Filters**

The following filters are provided for detailed EV analysis:

**Brand Filter**

Enables selection of EV manufacturers such as:

- Mercedes-Benz
- Nissan
- Porsche
- Tesla
- Volvo
- Audi
- BMW
- Tata, etc.

**Powertrain Filter**

Allows analysis based on drivetrain configuration:

- All
- AWD (All-Wheel Drive)
- FWD (Front-Wheel Drive)
- RWD (Rear-Wheel Drive)

**Car Filter**

Allows users to select individual EV models, for example:

- Audi e-Tron
- BMW iX
- Tata Nexon EV
- Porsche Taycan
- Tesla Model 3
- Hyundai Kona Electric, etc.

These filters enable detailed analysis of electric vehicles based on brand, model, and drivetrain type while minimizing the amount of data rendered.

**3. Number of Calculation Fields**

In Tableau, a calculated field is a user-defined field created using formulas rather than directly using raw data. Excessive calculated fields may affect dashboard performance; therefore, monitoring their number is important.

**Steps to Create a Calculated Field**

1. Right-click inside the **Data Pane**.
2. Select **Create → Calculated Field.**
3. Enter the field name.
4. Write the formula.
5. Click **OK**.

**Calculated Fields Used**

**1. Count**

This calculated field counts the number of car models.

**Formula:**

COUNT([Car])

**Purpose:**
Returns the total number of EV car records available in the dataset.

**2. Count_PowerTrain**

This calculated field counts the number of powertrain entries.

**Formula:**

COUNT([PowerTrain])

**Purpose:**
Returns the total number of drivetrain records (AWD, FWD, RWD, etc.) present in the dataset.